

# Mukesh Mani Tripathi

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## Summary

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AI and Embedded Systems Engineer with experience in robotics, embodied AI, computer vision, and automotive embedded software. Previously worked at RFA Engineering on embedded control systems for John Deere machinery using C/C++, MATLAB, and hardware-in-the-loop testing. Currently conducting research at Towson University on robotic repair systems, perception pipelines, and vision-language-action frameworks for manipulation tasks.

## Education

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| <b>Towson University</b> , Towson, MD<br>Master of Science in Computer Science (Cybersecurity)          | Expected May 2027 |
| <b>University of Wisconsin–Platteville</b> , Platteville, WI<br>Bachelor of Science in Computer Science | May 2023          |
| <b>North Lake College</b> , Irving, TX<br>Associate of Science                                          | May 2021          |

## Professional Experience

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**Research Assistant – Towson University, School of Emerging Technologies** Jan 2025 – Present

- Conduct research on embodied AI systems for robotic repair and manipulation tasks.
- Developed computer vision pipelines for structural damage detection and robotic perception using PyTorch and OpenCV.
- Built synthetic datasets and automated annotation workflows for robotic repair planning.
- Integrated perception outputs with robotic simulation environments for manipulation and task execution.
- Worked on multimodal vision-language-action frameworks for robotic furniture repair research.

**Application Development Intern – Erewhon Life Sciences** May 2026 – Aug 2026  
Remote, Baltimore, Maryland

- Assisted in the design, development, testing, and maintenance of software applications for healthcare and life science projects.
- Collaborated with development teams on feature implementation, debugging, and application testing.
- Wrote maintainable application code following software engineering best practices.
- Participated in technical reviews, troubleshooting, and project progress meetings.

**Embedded Software Engineer – RFA Engineering (Client: John Deere)** Jun 2023 – Dec 2024

- Developed embedded control software for construction machinery using C, C++, MATLAB, and Simulink.
- Implemented Blade Control, Grade Control, and IMU calibration features for machine automation systems.
- Built and maintained Jenkins CI pipelines for automated software builds and validation.
- Performed debugging and validation using hardware-in-the-loop systems and vehicle simulators.
- Supported migration of embedded software pipelines across infrastructure environments.

## Research Publications

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- Tripathi, M. M., Xin, H., Li, Q. *Repair2Skill: A Vision-Language-Action Framework for Robotic Furniture Repair*, AAAI Spring Symposium, April 2026. Affiliation: Towson University. (To appear)

## Technical Skills

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|------------------|-------------------------------------------------------------------|
| Programming      | Python, C, C++, MATLAB, JavaScript, Bash                          |
| AI / Robotics    | PyTorch, OpenCV, Computer Vision, Embodied AI, Robotic Simulation |
| Embedded Systems | CAN, J1939, SPI, I2C, MQTT, Hardware-in-the-loop Testing          |
| Tools            | Git, GitHub, Jenkins, CMake, GNU Make, Linux                      |
| Cybersecurity    | Wireshark, Nmap, Nessus, Burp Suite, OpenVAS                      |

## Projects

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### **Embedded32 – Open Automotive Embedded OS and J1939 Simulation Platform**

- Developed an open embedded platform for CAN and SAE J1939 simulation and diagnostics.
- Implemented scheduler, timers, message bus, diagnostics, and networking modules.
- Built transport protocol support including BAM and RTS/CTS communication.
- Developed ECU simulation tools and automated testing utilities.

### **Repair2Skill – Vision-Language-Action Robotic Repair Framework**

- Developed a robotic repair framework integrating vision, language reasoning, and manipulation planning.
- Built synthetic repair datasets and perception pipelines for damaged furniture detection.
- Integrated robotic simulation workflows for repair task execution and evaluation.

### **AI Recipe Recommendation System**

- Built a recommendation engine using TF-IDF and cosine similarity for ingredient-based recipe matching.
- Developed REST APIs using Flask for recommendation serving and dataset processing.

## Certifications

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- Google Cybersecurity Certificate
- CompTIA Security+ (In Progress)